Concept Quiz: Practice Sets 1-8

CFA Level II · 249 one-sentence questions · answers are on the last pages

How to use: read a question, answer it in your head, then check the answer pages (cover them while you read). Mark the box if you got it right. ★ = a concept you missed in practice sets 1-3 or Mock 10 (22 questions): review these first.

Ethics (21 questions)

ETH 1 A CFO tells a few analysts a nonpublic earnings warning: may you trade on it?

ETH 2 Which Standard requires fair and equitable dissemination of a recommendation to all clients?

ETH 3 Under VII(B), what may and may not be said about the CFA designation?

ETH 4 ★ A covered company offers to pay your commercial flights to its conference: allowed?

ETH 5 ★ When is company-provided travel acceptable under I(B)?

ETH 6 ★ A report presents an uncertain forecast as certain: which Standard is violated?

ETH 7 Which procedures does VI(B) Priority of Transactions recommend for personal trades?

ETH 8 If local law and the Standards differ, which applies under I(A)?

ETH 9 Whose information is protected by III(E) Preservation of Confidentiality?

ETH 10 A supervisor suspects violations by an employee: what is the appropriate response under IV(C)?

ETH 11 May a member use third-party research under V(A)?

ETH 12 What does the mosaic theory allow?

ETH 13 A personal buy order of four times the average volume to attract a merger partner is which violation?

ETH 14 What control must a firm use while it holds MNPI about an issuer?

ETH 15 Which fee arrangement for issuer-paid research best supports independence?

ETH 16 What does VI(C) require for referral fees?

ETH 17 Must a member tell clients about possible sanctions while a PCP investigation is pending?

ETH 18 Before recommending an asset allocation, what must be gathered to meet III(C) Suitability?

ETH 19 A new law takes effect before the firm updates its policies: what is the best fix?

ETH 20 Is a firm ban on staff investing in private placements consistent with the Standards?

ETH 21 Which additional compensation needs the employer's written consent under IV(B)?

FSA (41 questions)

FSA 1 A US company holds a euro receivable and the euro appreciates: gain or loss?

FSA 2 What mainly determines a subsidiary's functional currency?

FSA 3 Temporal method: where does the remeasurement gain or loss go?

FSA 4 Current rate method: where does the translation adjustment go?

FSA 5 Net asset exposure and the foreign currency depreciates: what happens to CTA?

FSA 6 Net monetary liability exposure under the temporal method and the foreign currency appreciates: gain or loss?

FSA 7 Which translation method gives the higher gross margin when the foreign currency depreciates?

FSA 8 In rising prices, which inventory method gives the higher gross margin?

FSA 9 How can a parent reduce balance sheet exposure of a subsidiary with positive net assets?

FSA 10 Proportionate consolidation vs equity method: what is the same, what is higher?

FSA 11 Rank total assets under acquisition, proportionate consolidation and equity method.

FSA 12 What income does an FVPL equity investment generate?

FSA 13 Equity method investment account roll-forward?

FSA 14 What happens to unrealized profit on an intercompany sale under the equity method?

FSA 15 Full vs partial goodwill: which gives the higher ROE and ROA?

FSA 16 Full goodwill formula?

FSA 17 US GAAP interest cost vs IFRS net interest on a pension plan?

FSA 18 Which pension items sit in P&L under IFRS?

FSA 19 Which assumption change lowers US GAAP pension cost?

FSA 20 A windfall tax benefit on stock compensation: effect on ETR under US GAAP vs IFRS?

FSA 21 Higher expected volatility at grant: effect on option expense, RSU expense and net income?

FSA 22 Higher expected dividend yield: effect on option expense and net income?

FSA 23 How are RSUs treated in the treasury stock method?

FSA 24 Loss ratio and expense ratio for a P&C insurer?

FSA 25 What does a combined ratio below 100% mean?

FSA 26 After years of weak pricing and capital loss, which phase of the P&C cycle follows?

FSA 27 Allowance for loan losses ÷ net charge-offs rising signals what?

FSA 28 What is in Common Equity Tier 1?

FSA 29 Effect of an inverted yield curve on a bank's net interest revenue?

FSA 30 A Beneish M-score above −1.78 signals what?

FSA 31 What are discretionary accruals?

FSA 32 Receivables growing faster than revenue and net income above CFO: what does it suggest?

FSA 33 Segment capex share ÷ segment asset share above 1: what does it suggest?

FSA 34 Reclassifying interest paid from CFF to CFO: effect on CFO and CFF?

FSA 35 Five-step DuPont formula for ROE?

FSA 36 ★ How do you find financial leverage from revenue, asset turnover and liabilities?

FSA 37 ★ In a table with columns 'Most recent year, Previous year, Two years ago', how do you read a trend?

FSA 38 Depreciable lives longer than peers: what happens to D/E after adjusting to the shorter peer lives?

FSA 39 CFO before interest and taxes ÷ operating income well above 1 indicates what?

FSA 40 Carrying basis under IFRS 9 for amortized cost, FVOCI and FVPL?

FSA 41 An SPE funded by an outside bank is consolidated: effect on consolidated cash and debt?

Equity (32 questions)

EQ 1 ★ FCFE formula with a target debt ratio?

EQ 2 ★ What is excluded from working capital investment for FCFE?

EQ 3 When is FCFE preferred over a dividend discount model?

EQ 4 Residual income formula?

EQ 5 Residual income model value?

EQ 6 Market value added (MVA)?

EQ 7 EVA formula?

EQ 8 Compare NOPAT with what, and net income with what?

EQ 9 Single-stage RI model value?

EQ 10 Justified P/B in the RI framework?

EQ 11 If ROE equals the cost of equity, what is the intrinsic value?

EQ 12 What does a persistence factor of 1 mean, and near 0?

EQ 13 Which companies tend to have a low persistence factor?

EQ 14 Terminal value in a multistage RI model with persistence factor ω?

EQ 15 If the terminal price is given (or price = book at the horizon), what is the terminal term?

EQ 16 Clean surplus relation?

EQ 17 Which items violate clean surplus?

EQ 18 Does the RI model recognize value earlier or later than a DDM?

EQ 19 Non-recurring items in RI valuation: adjust income or book value?

EQ 20 Value of a perpetual preferred share?

EQ 21 ★ DLOC from a control premium?

EQ 22 ★ Total discount from DLOC and DLOM?

EQ 23 Mean or median P/E for comparables with an outlier?

EQ 24 Capitalized cash flow method (CCM) formula?

EQ 25 Excess earnings method (EEM): what are the components?

EQ 26 Excess earnings in EEM?

EQ 27 The three approaches to private company valuation?

EQ 28 ★ Synergistic or financial control premium when the target is combined with another portfolio company?

EQ 29 What does normalizing earnings mean?

EQ 30 An owner is paid less than the market rate: normalized EBITDA goes up or down?

EQ 31 Trailing P/E denominator vs forward vs normalized?

EQ 32 Sustainable growth rate?

Fixed Income (44 questions)

FI 1 When can OAS of two bonds be compared, and which is cheaper?

FI 2 Are the rates in a binomial tree forecasts?

FI 3 Inputs to calibrate an interest rate tree?

FI 4 Does volatility change the tree value of an option-free bond?

FI 5 Callable and putable bond values in terms of a straight bond?

FI 6 Volatility rises: effect on callable, putable, and their OAS?

FI 7 Rates fall sharply: what happens to a callable bond's duration and convexity?

FI 8 What is the effective duration of a floating-rate note?

FI 9 Upward-sloping curve flattens (level unchanged): effect on the call option value?

FI 10 Conversion price vs market conversion price?

FI 11 Minimum value of a convertible bond?

FI 12 Conversion price above the share price: what does the convertible behave like?

FI 13 Dividend above the threshold: effect on the conversion ratio?

FI 14 Value of a callable, putable convertible?

FI 15 Pathwise valuation?

FI 16 Forward rate f(2,1) from spot rates?

FI 17 Forward price of a zero F(j,k)?

FI 18 Rolling down the curve: how is the return computed?

FI 19 Projected spot curve below the forward curve: bond valued how?

FI 20 What does the dominance principle say?

FI 21 Bullish steepening, bullish flattening, bearish flattening?

FI 22 Swap spread, I-spread, Z-spread, TED spread?

FI 23 When is the swap curve a better benchmark than the government curve?

FI 24 Which YTM assumptions apply, and what if coupons are reinvested below the YTM?

FI 25 ★ Expected loss and loss given default?

FI 26 Effect of credit migration on expected return?

FI 27 Fair value of a risky bond and its credit spread?

FI 28 Constant hazard rate h: PD and survival probability?

FI 29 Lower risk-neutral PD: effect on CVA and credit spread?

FI 30 Structural model: how are equity and risky debt viewed?

FI 31 Weakness of a reduced-form model?

FI 32 A CDS coupon of 1% and a spread of 1.8%: who pays upfront?

FI 33 Protection seller: when does it gain?

FI 34 Bond spread above CDS spread: what is the basis trade?

FI 35 Expecting the CDS curve to steepen: which trade?

FI 36 Economy improves: which index CDS trade?

FI 37 Expected leveraged buyout financed with debt: stock and CDS trade?

FI 38 Which pool analysis fits short-term, homogeneous, granular pools? Medium-term? Non-granular?

FI 39 Rating of subordinated debt vs the issuer's senior unsecured rating?

FI 40 Credit term structure for the highest-rated vs lower-rated issuers?

FI 41 Pure expectations, local expectations, liquidity preference, segmented markets, preferred habitat?

FI 42 Which short-rate models can go negative?

FI 43 What drives short-term vs long-term rate volatility?

FI 44 Which key rate duration is highest for an option-free bond?

Derivatives (22 questions)

DER 1 ★ Value of a long forward on a non-dividend stock after initiation?

DER 2 Forward price with a dividend?

DER 3 A forward is overpriced versus the model: which arbitrage?

DER 4 A futures price is below the model price: which arbitrage?

DER 5 Treasury futures equilibrium price?

DER 6 Accrued interest on a bond?

DER 7 Fixed rate of a swap or equity swap from discount factors?

DER 8 Value of a pay-fixed swap after initiation?

DER 9 Currency swap valuation steps?

DER 10 How do you turn fixed-rate debt into floating-rate debt?

DER 11 Which instrument gains if rates rise but limits the loss if they do not?

DER 12 Signs of delta, gamma, vega, theta, rho for a long call? For a long put?

DER 13 What does a short option position have: gamma and theta signs?

DER 14 Option value one day later with daily theta −0.021?

DER 15 Delta hedge of a short call position on 1,200 shares?

DER 16 Is early exercise ever optimal for an American call on a non-dividend stock?

DER 17 ★ How do you value an American put on a binomial tree?

DER 18 Risk-neutral probability of an up move?

DER 19 What is implied volatility?

DER 20 Volatility smile, skew, surface?

DER 21 Futures price of a commodity with storage costs and convenience yield?

DER 22 A client owning shares buys puts: change in delta, gamma, vega?

Quant (25 questions)

QNT 1 ★ Adjusted R^2 formula and its special property?

QNT 2 Standard error of estimate?

QNT 3 Overall F-statistic and nested F-statistic?

QNT 4 ★ AIC vs BIC: purpose and which value is preferred?

QNT 5 Significant Breusch–Pagan test: problem, effect and fix?

QNT 6 Durbin–Watson below the lower critical value: what does it show and what is the fix?

QNT 7 Signals of multicollinearity and how to detect it?

QNT 8 Durbin–Watson zones?

QNT 9 Mean-reverting level of an AR(1) model?

QNT 10 How do you test for seasonality in quarterly and monthly data?

QNT 11 Covariance stationarity requires what?

QNT 12 Dickey–Fuller null hypothesis and what to do if not rejected?

QNT 13 Regressing one unit-root series on another: when valid?

QNT 14 ★ A random walk as an AR(1): coefficients and best forecast?

QNT 15 ARCH(1) test: which coefficient matters?

QNT 16 Log-linear trend forecast?

QNT 17 LASSO, SVM, CART, k-means, PCA: one line each?

QNT 18 Agglomerative vs divisive clustering?

QNT 19 What is a deep learning network?

QNT 20 Overfitting: which learning and the learning curve axes?

QNT 21 Text cleansing vs text wrangling?

QNT 22 Stemming vs lemmatization?

QNT 23 Steps of a structured data project in order?

QNT 24 Winsorization vs trimming; normalization vs standardization?

QNT 25 Order of the harmonic, geometric and arithmetic means?

Portfolio Management (26 questions)

PM 1 Information ratio formula and its components?

PM 2 IR of an index fund that meets its objective exactly?

PM 3 Does changing the benchmark change the Sharpe ratio or the information ratio?

PM 4 Does scaling all active weights change the IR?

PM 5 What does a closet index fund look like?

PM 6 Active return vs alpha?

PM 7 Security selection and asset allocation formulas?

PM 8 IC when forecasts are correct a share p of the time?

PM 9 Breadth with correlated decisions?

PM 10 Expected active return from IC, breadth and active risk?

PM 11 Optimal active risk and maximum Sharpe ratio with a transfer coefficient?

PM 12 Weight in the active portfolio?

PM 13 Does adding cash to an active portfolio change the Sharpe ratio or the IR?

PM 14 Macroeconomic, fundamental and statistical factor models?

PM 15 What does APT assume and not specify?

PM 16 Active return from factor tilts?

PM 17 ★ What does a style read from active sensitivities mean: size +, value −?

PM 18 Contribution of a factor to active risk squared?

PM 19 Parametric VaR: formula and limits?

PM 20 Number of distinct covariances for n factors?

PM 21 ETF trading above NAV: what do authorized participants do?

PM 22 ETF uses: liquidity management, rebalancing, completion?

PM 23 Fundamental asset pricing equation?

PM 24 ★ Real rates depend on what?

PM 25 Positive output gap means what?

PM 26 ★ Rents: which part is in the discount rate and which in the cash flow?

Economics (14 questions)

ECO 1 Dornbusch model: how does the currency respond to monetary expansion?

ECO 2 Portfolio balance approach: effect of sustained deficits financed by debt?

ECO 3 Mundell–Fleming: expansionary monetary and restrictive fiscal policy with high capital mobility?

ECO 4 Mundell–Fleming: restrictive monetary and fiscal policy with low capital mobility?

ECO 5 Which indicators signal a currency crisis?

ECO 6 What is a carry trade and when does it lose?

ECO 7 Under uncovered interest parity, what is the high-yield currency expected to do?

ECO 8 All-in forward rate from forward points?

ECO 9 Mark-to-market of a forward position?

ECO 10 The CAD depreciates against the USD: what happens to the USD/CAD quote?

ECO 11 Relative PPP: does it hold in the short run or the long run?

ECO 12 Neoclassical growth: steady-state growth per capita?

ECO 13 Neoclassical model and capital flows?

ECO 14 Bid cross rate EUR/AUD from EUR/USD and AUD/USD quotes?

Corporate Issuers (15 questions)

CORP 1 Special dividend vs higher regular dividend?

CORP 2 Stable vs constant payout dividend policy?

CORP 3 Debt-financed repurchase: when does EPS rise?

CORP 4 Cash buyback: effect on assets, equity and leverage?

CORP 5 Price after a stock dividend?

CORP 6 What is greenmail?

CORP 7 Repurchase methods and prices?

CORP 8 Reasons for a low payout ratio?

CORP 9 Which items raise operating income in a merger?

CORP 10 Which CAPM for a private domestic company and for a subsidiary in an emerging country?

CORP 11 Pre-tax cost of debt for a company with no traded bonds?

CORP 12 Cross-holdings, pyramids, executives on each other's boards?

CORP 13 Controlling votes far above ownership: which problem and which remedy?

CORP 14 ESG data: vendors vs company-reported?

CORP 15 Why can a green bond have a tighter spread than a conventional bond of the same issuer?

Alternative Investments (9 questions)

ALT 1 REIT vs REOC?

ALT 2 ★ NAVPS method?

ALT 3 A highly leveraged REIT trading far below NAVPS: what is the concern?

ALT 4 Dedicated short vs short-biased fund?

ALT 5 What makes a life settlement attractive?

ALT 6 ★ Real estate cycle order?

ALT 7 ★ How do you compare the effect of unreimbursed expenses across properties?

ALT 8 Direct capitalization for a property with temporarily disrupted income?

ALT 9 Commodity in backwardation: relation of convenience yield and storage costs?

Answers

Same numbering as the questions. ★ = concept missed before.

Ethics

ETH 1 No. Selective disclosure does not make it public; it is MNPI under II(A). Encourage disclosure and do not act.

ETH 2 III(B) Fair Dealing.

ETH 3 May say 'I am a CFA charterholder'; may not imply superior skill or promise better returns.

ETH 4 ★ No (I(B)). Pay your own or your firm's travel. Only a token gift (a branded mug) is fine.

ETH 5 ★ Only for a remote, hard-to-reach site with no commercial transport; reasonable travel, not lavish.

ETH 6 ★ V(B) Communication with Clients: failure to separate fact from opinion and to disclose uncertainty.

ETH 7 Pre-clearance, and duplicate trade confirmations or statements sent to the employer.

ETH 8 The stricter one.

ETH 9 Former, current and prospective clients.

ETH 10 Limit the employee's trading and increase monitoring while the facts are investigated.

ETH 11 Yes, but with reasonable efforts to check it is sound (due diligence).

ETH 12 Combining public information and nonmaterial nonpublic information.

ETH 13 II(B) Market Manipulation.

ETH 14 Heightened review of trading (restricted or watch list); it need not stop all trading.

ETH 15 A disclosed flat fee, with no bonus, regardless of the conclusion.

ETH 16 Disclose to the employer, clients and prospective clients any compensation or benefit paid or received for referrals.

ETH 17 No. Sanctions are not official until the process ends.

ETH 18 The client's profile: objectives, constraints, demographics and liquidity needs.

ETH 19 A process to learn about legal changes promptly, before they take effect.

ETH 20 Yes. VI(B) recommends strict limits; a full ban goes further but is allowed.

ETH 21 Both monetary and non-monetary compensation (cash and gifts).

FSA

FSA 1 Gain (a payable would give a loss).

FSA 2 Currency of revenues and operating costs (financing is secondary).

FSA 3 Net income.

FSA 4 OCI, accumulated as CTA in equity.

FSA 5 It decreases (it increases if the currency appreciates).

FSA 6 Loss in net income.

FSA 7 Current rate (temporal gives the higher margin when it appreciates).

FSA 8 FIFO (older, cheaper costs).

FSA 9 Reduce net assets, e.g. a dividend paid from local cash.

FSA 10 Net income and equity are the same; revenue, assets and liabilities are higher.

FSA 11 Acquisition > proportionate > equity method.

FSA 12 Holding gain or loss plus dividends received (not the investee's net income).

FSA 13 Cost + share of net income − share of dividends − amortization of the excess purchase price.

FSA 14 The investor's share is deferred until the goods are sold to third parties.

FSA 15 Partial goodwill (smaller assets and equity, same net income).

FSA 16 Purchase price ÷ % acquired − fair value of identifiable net assets.

FSA 17 US GAAP: beginning PBO × discount rate. IFRS: beginning funded status × discount rate.

FSA 18 Service cost (current and past) and net interest; remeasurements go to OCI.

FSA 19 Lower estimated salary increases (a lower expected return on assets raises cost).

FSA 20 US GAAP: lower effective tax rate. IFRS: excess goes to equity, no effect on ETR.

FSA 21 Option fair value and expense rise, net income falls; RSUs are unaffected.

FSA 22 Option value and expense fall; net income rises.

FSA 23 No exercise proceeds; dilutive shares = number − unrecognized expense ÷ average price.

FSA 24 Loss ratio = (claims paid + change in loss reserves) ÷ net premiums earned; expense ratio = underwriting expenses ÷ net premiums written.

FSA 25 An underwriting profit (loss ratio + expense ratio < 100%).

FSA 26 A hard market: less competition, higher premiums.

FSA 27 A larger cushion, more conservative provisioning (provision ÷ charge-offs > 1 is conservative).

FSA 28 Common stock, APIC, retained earnings, OCI, less intangibles and DTAs (preferred stock is Additional Tier 1).

FSA 29 Falls (short-term funding costs rise relative to long-term lending rates).

FSA 30 A higher probability of earnings manipulation.

FSA 31 The residual of the accruals regression (non-discretionary accruals are explained by business activity).

FSA 32 Aggressive revenue recognition (low earnings quality).

FSA 33 The segment is being grown (invested in more than its asset share).

FSA 34 CFO lower, CFF higher.

FSA 35 Tax burden × interest burden × EBIT margin × asset turnover × financial leverage.

FSA 36 ★ Assets = revenue ÷ turnover; equity = assets − liabilities; leverage = assets ÷ equity.

FSA 37 ★ Time runs right to left: the first column is the newest year. Check the column order before reading a trend.

FSA 38 Depreciation up, net income and equity down, so D/E rises.

FSA 39 Earnings are backed by cash flow.

FSA 40 Amortized cost: historical cost. FVOCI and FVPL: fair value.

FSA 41 Both increase.

Equity

EQ 1 ★ NI − (1 − DR)(FCInv − Dep) − (1 − DR)(WCInv).

EQ 2 ★ Cash and cash equivalents: WCInv = Δ(current assets excl. cash) − Δ(current liabilities).

EQ 3 Dividends differ from capacity to pay, no dividends, or a control perspective.

EQ 4 RI = net income − cost of equity × beginning book value of equity.

EQ 5 V0 = B0 + present value of expected residual income.

EQ 6 Market value of capital − book value of total capital (equity-only: V0 − B0).

EQ 7 NOPAT − WACC × invested capital.

EQ 8 NOPAT (before interest) with WACC; net income (after interest) with the cost of equity.

EQ 9 V0 = B0 + B0(ROE − r)/(r − g).

EQ 10 P/B = 1 + (ROE − r)/(r − g) = (ROE − g)/(r − g).

EQ 11 Book value (no residual income).

EQ 12 1: residual income stays constant forever. Near 0: it vanishes right after the horizon.

EQ 13 Extreme ROE, high accruals, special (non-recurring) items.

EQ 14 RI(T+1) ÷ (1 + r − ω), discounted back from T.

EQ 15 PV of (P_T − B_T); it is zero when price equals book. No persistence factor is needed.

EQ 16 B1 = B0 + EPS − DPS (no other changes in equity).

EQ 17 OCI items such as unrealized AFS gains and currency translation adjustments.

EQ 18 Earlier (a large part comes from current book value).

EQ 19 Adjust income, not book value.

EQ 20 D ÷ r.

EQ 21 ★ DLOC = 1 − 1/(1 + control premium). Convert a premium to a discount first.

EQ 22 ★ 1 − (1 − DLOC)(1 − DLOM): they multiply, not add.

EQ 23 Median.

EQ 24 Value = FCFF1 ÷ (WACC − g), or FCFE1 ÷ (r − g); use next year's cash flow.

EQ 25 Working capital + fixed assets (tangible) + capitalized excess earnings (intangibles).

EQ 26 Normalized earnings − required return on working capital and fixed assets.

EQ 27 Income (DCF, CCM, EEM), market (GPCM, GTM, PTM), asset-based.

EQ 28 ★ Synergistic (higher than a financial premium).

EQ 29 Adjusting revenues and costs (non-market owner pay, personal expenses, non-recurring items) so the company can be compared with public peers.

EQ 30 Down, by the gap (costs are understated).

EQ 31 Trailing: last four quarters' EPS. Forward: expected EPS. Normalized: average over a business cycle.

EQ 32 g = retention ratio × ROE.

Fixed Income

FI 1 Same credit rating and same tree; the larger OAS is cheaper.

FI 2 No. They are stochastic paths calibrated to today's curve and the volatility assumption.

FI 3 Volatility, an interest rate model, and market values of benchmark bonds.

FI 4 No: same as the spot rate value.

FI 5 Callable = straight − call; putable = straight + put.

FI 6 Callable value falls, putable rises; OAS of the callable falls, OAS of the putable rises.

FI 7 Duration falls and convexity turns negative.

FI 8 About the time to the next reset.

FI 9 Call value rises (forward rates fall).

FI 10 Conversion price = par ÷ conversion ratio; market conversion price = market price ÷ conversion ratio.

FI 11 The greater of conversion value and straight value.

FI 12 A bond (driven by rates and credit spread).

FI 13 It increases (conversion price falls).

FI 14 Straight + call on stock − call on bond + put on bond.

FI 15 Discount each cash flow by the product of the one-period rates along the path.

FI 16 $(1 + s3)^3 \div (1 + s2)^2 - 1$.

FI 17 $1 \div (1 + f(j,k))^k$; the exponent is the length of the forward period.

FI 18 Buy price from today's spot, sell price from the shorter-maturity spot (curve unchanged); annualize over the holding period.

FI 19 Undervalued: buy.

FI 20 Buy the asset with the lower price per unit of payoff and sell the other.

FI 21 Bullish = yields fall. Steepening: short yields fall more (recession). Flattening: long yields fall more (flight to quality, extend duration with a barbell). Bearish flattening: short yields rise more (expansion).

FI 22 Swap spread = swap rate − government yield; I-spread = bond yield − swap rate; Z-spread: constant spread on spot rates to match price; TED = 3-month MRR − T-bill yield.

FI 23 When government bonds are illiquid.

FI 24 Held to maturity, no default, reinvestment at the YTM; a lower reinvestment rate gives a realized return below the YTM.

FI 25 ★ Expected loss = PD × LGD; LGD = 1 − recovery rate.

FI 26 Negative: downgrades widen spreads more than upgrades narrow them.

FI 27 Fair value = VND − CVA; spread = YTM(fair value) − benchmark YTM.

FI 28 $PD_t = PS(t-1) \times h$; $PS_t = (1 - h)^t$. Varying hazard rates: $PS_t = \Pi(1 - h_i)$.

FI 29 CVA falls, fair value rises, spread falls.

FI 30 Equity = call on the assets; risky debt = riskless debt − put on the assets.

FI 31 It does not explain why default happens (default is a surprise); its strength is that assets need not trade.

FI 32 The protection buyer. Upfront ≈ (spread − coupon) × CDS duration; price = 100 − upfront.

FI 33 When the spread narrows below the original spread (loss if it widens).

FI 34 Buy the bond and buy protection.

FI 35 Sell protection at the short end, buy protection at the long end.

FI 36 Sell high-yield protection, buy investment-grade protection.

FI 37 Buy the shares and buy CDS protection.

FI 38 Statistical; portfolio; loan-by-loan.

FI 39 Notched below.

FI 40 Highest-rated: flat or slightly upward. Lower-rated: steeper; spreads narrow when growth expectations are strong.

FI 41 No premium; same short-holding return; premium rising with maturity; never leave a segment; leave if paid enough.

FI 42 Vasicek and Ho-Lee can; CIR ($\sigma\sqrt{r}$) and KWF (models $\ln r$) cannot. Ho-Lee and KWF are arbitrage-free.

FI 43 Short-term: monetary policy. Long-term: economy and inflation expectations.

FI 44 The one at maturity.

Derivatives

DER 1 ★ $S_t - F0 \div (1 + r)^{(T-t)}$.

DER 2 $S0(1 + r)^T - FV$ of the dividends (compounded to expiration).

DER 3 Sell the forward, borrow, buy the asset (cash-and-carry).

DER 4 Buy the futures, short the bond, lend (reverse cash-and-carry).

DER 5 $[(S + AI0)(1 + r)^T - AI_T - FV(\text{coupons})] \div$ conversion factor.

DER 6 Days since the last coupon ÷ days in the coupon period × the coupon.

DER 7 $(1 - Z_n) \div \Sigma Z$ (times periods per year to annualize).

DER 8 $\Sigma Z \times$ (new fixed rate − old fixed rate) × notional.

DER 9 Notionals at the initial spot; PV each leg in its own currency's factors; convert at today's spot.

DER 10 Receive fixed and pay floating.

DER 11 A payer swaption.

DER 12 Call: +, +, +, −, +. Put: −, +, +, −, −. A short option reverses the signs.

DER 13 Negative gamma, positive theta.

DER 14 Value today + theta (value falls by 0.021); not gamma.

DER 15 Buy delta × 1,200 shares.

DER 16 No; it equals the European call.

DER 17 ★ Start at maturity and work back; at every node take the larger of the continuation value and the exercise value (K − S).

DER 18 q = (1 + r − d) ÷ (u − d).

DER 19 The volatility backed out of market option prices; forward-looking.

DER 20 Smile: higher for ITM and OTM than ATM. Skew: varies with strike in one direction. Surface: across strikes and maturities.

DER 21 Spot + storage costs − convenience yield; backwardation when convenience yield > storage cost.

DER 22 Delta lower, gamma positive, vega positive.

Quant

QNT 1 ★ 1 − [(n − 1)/(n − k − 1)](1 − R^2); it can fall when a variable is added and can be negative.

QNT 2 √[SSE ÷ (n − k − 1)].

QNT 3 (RSS/k) ÷ (SSE/(n − k − 1)); [(SSE_R − SSE_U)/q] ÷ [SSE_U/(n − k − 1)].

QNT 4 ★ AIC for forecasting, BIC for goodness of fit; the lower value (more negative) is better for both.

QNT 5 Conditional heteroskedasticity: standard errors too low, t too high; use White robust errors.

QNT 6 Positive serial correlation; use Newey–West errors.

QNT 7 High R^2, significant F, insignificant t-statistics; check VIF; drop or combine variables.

QNT 8 0 to dl: positive serial correlation; dl to du: inconclusive; du to 4 − du: none; then inconclusive; 4 − dl to 4: negative.

QNT 9 b0 ÷ (1 − b1); the series moves toward it.

QNT 10 Check the autocorrelation of residuals at lag 4 (quarterly) or lag 12 (monthly); add that lag if significant.

QNT 11 Constant and finite mean, variance and covariance with own lags.

QNT 12 Unit root; first-difference the series.

QNT 13 Valid only if the series are cointegrated (or neither has a unit root).

QNT 14 ★ b0 ≈ 0, b1 ≈ 1; best forecast is the current value; no mean reversion.

QNT 15 Regress squared residuals on lagged squared residuals; c1 significantly different from 0 means ARCH.

QNT 16 ln(y) = b0 + b1·t, so y = e^(b0 + b1·t).

QNT 17 LASSO: penalized regression, feature selection; SVM: widest-margin linear classifier; CART: non-linear classification and regression; k-means: k chosen in advance; PCA: reduces features.

QNT 18 Agglomerative: bottom-up (merge). Divisive: top-down (split).

QNT 19 A neural network with at least two hidden layers.

QNT 20 Supervised learning; accuracy in validation/test sample vs size of the training sample.

QNT 21 Cleansing: remove HTML, punctuation, numbers, white space. Wrangling: lowercase, stop words, stemming, lemmatization.

QNT 22 Stemming cuts words to a base (may not be a word); lemmatization uses a dictionary (better, costlier).

QNT 23 Conceptualization, data collection, preparation and wrangling, exploration, model training.

QNT 24 Winsorize replaces outliers with a percentile value; trimming removes them. Normalization rescales to 0–1 (outlier sensitive); standardization centers at 0 with st. dev. 1.

QNT 25 Harmonic < geometric < arithmetic (equal only if all values are equal).

Portfolio Management

PM 1 IR = active return ÷ active risk = TC × IC × √BR.

PM 2 Zero.

PM 3 Only the information ratio.

PM 4 No.

PM 5 Low active risk, Sharpe ratio close to the benchmark.

PM 6 Active return = portfolio − benchmark return; alpha is the return beyond what risk exposures explain.

PM 7 Selection = Σ w_p(R_p − R_b); allocation = Σ R_b(w_p − w_b).

PM 8 IC = 2p − 1.

PM 9 BR = N ÷ [1 + (N − 1)r].

PM 10 E(R_A) = IC × √BR × σ_A.

PM 11 σ_A* = TC × (IR ÷ SR_B) × σ_B; SR_P = √(SR_B² + TC²·IR²).

PM 12 Optimal active risk ÷ active risk of the active portfolio.

PM 13 IR changes; the Sharpe ratio does not.

PM 14 Macro: factors are surprises, intercept = expected return; fundamental: attributes, sensitivities first; statistical: PCA, hard to interpret.

PM 15 Returns follow a factor model, specific risk is diversifiable, no arbitrage; it does not specify the number or identity of the factors.

PM 16 Σ (fund − benchmark sensitivity) × factor return; selection = active return − tilts.

PM 17 ★ Small-cap and growth.

PM 18 Factor's active risk (%²) ÷ total active risk squared; active specific risk is the rest.

PM 19 $(z \times \sigma - \mu) \times$ value, scaled by $\sqrt{\text{time}}$; ignores liquidity; high kurtosis means normal VaR understates tail risk.

PM 20 $n(n-1)/2$.

PM 21 Create ETF shares by delivering the basket and sell the ETF shares.

PM 22 Equitize cash quickly; move to target weights cheaply; fill temporary gaps in the allocation.

PM 23 Price = E(CF) ÷ (1 + Rf + π + θ + φ).

PM 24 ★ Higher with higher expected real growth and with higher volatility of real growth; compare real growth (nominal − inflation) first.

PM 25 Actual GDP above potential: inflation pressure, low unemployment.

PM 26 ★ Tenant credit risk is in the discount rate (φ); rent growth is in the cash flows.

Economics

ECO 1 It overshoots (depreciates more than the long run) because prices are sticky.

ECO 2 Currency depreciates in the long run.

ECO 3 Currency depreciates.

ECO 4 Currency appreciates.

ECO 5 CRRIB + M2: capital inflows surge, reserves fall, real exchange rate overvalued, inflation up, banking stress, M2/reserves up.

ECO 6 Borrow the low-yield currency, invest in the high-yield one; it loses if the funding currency appreciates (UIRP failing).

ECO 7 Depreciate by the rate difference.

ECO 8 Spot + points ÷ 10,000.

ECO 9 Compare with the current forward rate for the remaining term, then discount to today.

ECO 10 It falls.

ECO 11 Better in the long run; poor in the short run.

ECO 12 $\theta \div (1 - \alpha)$; capital deepening has diminishing returns.

ECO 13 Capital flows from rich to poor countries, helping convergence.

ECO 14 Bid EUR/USD × 1 ÷ offer AUD/USD.

Corporate Issuers

CORP 1 Special: one-time, no ongoing commitment. Regular increase: permanent commitment (a positive signal).

CORP 2 Stable: same amount (or smooth growth). Constant payout: same percentage of earnings.

CORP 3 When the earnings yield (E/P) exceeds the after-tax cost of debt.

CORP 4 Assets and equity fall; leverage ratios rise.

CORP 5 Old price ÷ (1 + stock dividend %).

CORP 6 Buying back a hostile bidder's shares at a premium to stop a takeover.

CORP 7 Open market: flexible. Tender or Dutch auction: premium. Negotiated: discount or premium (greenmail).

CORP 8 High growth opportunities, high flotation costs, high taxes on dividends.

CORP 9 Revenue synergies and cost synergies (dis-synergies and one-time costs lower it).

CORP 10 Expanded CAPM (size and company-specific premiums); extended/international CAPM with a country risk premium.

CORP 11 Interpolate comparable yields by maturity.

CORP 12 Horizontal structure, vertical structure, interlocking directorates.

CORP 13 Principal–principal conflict; a voting cap.

CORP 14 Vendors: subjective methods. Company-reported: inconsistent and voluntary.

CORP 15 Strong investor demand.

Alternative Investments

ALT 1 REIT: tax-exempt if it distributes most income, high yield, limited reinvestment. REOC: taxable, more flexibility.

ALT 2 ★ Value properties by direct capitalization (NOI ÷ cap rate), add other assets, subtract debt, divide by shares.

ALT 3 Refinancing risk (hard to raise capital).

ALT 4 Dedicated short: almost no long exposure. Short-biased: net short with some long positions.

ALT 5 Low surrender value, low premiums, earlier death than expected.

ALT 6 ★ Recovery, expansion, hyper-supply, recession; high vacancy means hyper-supply or recession, and recovery comes next.

ALT 7 ★ As a percentage of effective gross income, not in dollars.

ALT 8 Use a normalized (stabilized) NOI, or a DCF; not last year's NOI.

ALT 9 Convenience yield exceeds storage costs.